

SHAP Explainability Report and Analyst Review Template

NovaPay Fraud Detection

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1. Purpose

A fraud score on its own is not enough for a financial decision. Analysts need to know why a transaction was flagged, and regulators increasingly require that automated decisions be explainable. This report documents the explainability work carried out on the recommended Random Forest model using SHAP (SHapley Additive exPlanations), and proposes a standard review template so that every flag reaching a human carries its reasons, its key features, and a confidence score.

2. Method

SHAP was applied to the trained Random Forest model. Two levels of explanation were produced. The first is a global view, which measures how much each feature contributes to the model's predictions across the whole dataset, summarised through SHAP summary plots. The second is a local view, which decomposes a single prediction into the contribution of each feature, showing how much each one pushed that specific transaction toward or away from a fraud classification.

3. Global explanation: what drives the model overall

The global SHAP analysis, measured by mean absolute SHAP value, ranked the strongest contributors as follows: twenty-four-hour transaction velocity, IP risk score, one-hour transaction velocity, account age, velocity burst, internal risk score, the under-thirty-day account-age bucket, and device trust score, followed by transaction amount

and the various IP-risk and device-trust bucket flags. In plain terms, the model relies most on how rapidly an account is transacting, how risky its network origin is, how new the account is, and how trusted its device is. These match the signals identified during exploratory analysis, which is reassuring, because it shows the model is learning genuine fraud patterns rather than spurious correlations.

One important nuance is that several signals appear in more than one form. Account newness, for example, is represented by the raw account-age value, by an under-thirty-day bucket, and by a new-account flag, while IP risk and device trust each appear as a raw score and as bucketed flags. Because their influence is split across several encoded features, the true thematic importance of account newness, IP risk, and device trust is greater than any single bar suggests.

The directional view from the SHAP beeswarm plot confirms the relationships point the expected way. High transaction velocity, high IP risk, high internal risk, an active velocity burst, and membership of the under-thirty-day account bucket all push a transaction toward a fraud classification, while low device trust pushes toward fraud and high device trust pushes away from it. None of the directions are counterintuitive, which is a further sign the model is behaving sensibly.

3a. Engineered features behind the signals

Several of the most influential features were engineered rather than raw, which is worth recording because they encode domain intuition about fraud. A velocity ratio compares one-hour to twenty-four-hour activity, capturing how much faster an account is transacting now than its recent norm, since a fraudster typically bursts activity in a short window. A combined risk score blends IP risk, internal risk, and chargeback history into a single weighted measure. Two binary flags capture dangerous combinations: a high-risk combo where high IP risk, low device trust, and a velocity burst all fire together, which occurred in 574 transactions, and a new-account-high-velocity flag marking brand-new accounts transacting rapidly, the classic account-takeover pattern, which occurred in 616 transactions. One caution to note for future work is that the combined risk score

incorporates chargeback history, which can reflect information confirmed after a transaction, so it should be validated to ensure it only uses data available at the moment of scoring.

4. Local explanation: explaining a single decision

To show how the model explains an individual case, SHAP was applied to a flagged transaction, referred to as TXN_001. The model assigned it a fraud probability of 94%. The decomposition showed three features driving that decision: a one-hour velocity of seven transactions, which contributed the largest push toward fraud, an IP risk score of 0.95, and an account age of only five days. Each of these moved the prediction toward fraud, and together they explain the high confidence. This is the form of explanation that would accompany every flag in production, so that an analyst can see not only that a transaction was flagged but precisely why.

5. Why this matters

Explainability serves three purposes here. It builds analyst trust, because a flag that comes with clear reasons is far more actionable than an opaque score. It supports regulatory compliance, since frameworks such as GDPR Article 22 require that individuals can be given an explanation for automated decisions. And it acts as a safeguard, because inspecting the drivers of a model can reveal bias or data leakage before it reaches production.

6. Scope and limitations

This report covers the global and per-transaction SHAP analysis that was performed. It does not include a dedicated study of which features most often drive false positives, that is, an analysis isolating the legitimate transactions the model wrongly flagged and examining their SHAP profiles. On the recommended Random Forest model there were eleven such false positives in the test set, a small enough number that a focused SHAP study of that group is a natural and worthwhile next step. It would identify whether

particular features systematically push borderline-legitimate transactions over the threshold, and would help tune the review band. This is noted as future work rather than something completed here.

Analyst Review Template

The following template operationalises the SHAP output. It is designed so that every transaction routed to human review arrives with a confidence score, the top contributing features, and standardised reason codes, giving the analyst a consistent and explainable starting point.

Reason code dictionary

These short codes standardise the most common drivers, so flags can be logged, searched, and reported consistently.

Code	Meaning	Underlying feature
VEL-24H	Elevated transactions over 24 hours	txn_velocity_24h
VEL-1H	Rapid transactions in the last hour	txn_velocity_1h
VEL-BURST	Sudden burst of activity vs recent norm	velocity_burst / velocity_ratio
IP-RISK	High IP risk score	ip_risk_score
RISK-INT	High internal risk score	risk_score_internal
DEV-LOW	Low device trust	device_trust_score
ACCT-NEW	New or young account (under 30 days)	account_age_days / account_very_new
AMT-MID	Amount in elevated-risk band	amount_usd
COMBO	Multiple danger signals firing together	high_risk_combo

Confidence score and decision band

The confidence score is the model's predicted probability of fraud. It maps to a decision band, which determines whether the transaction is approved automatically, sent to review, or blocked.

Fraud probability	Band	Action
Below 0.30	Low	Approve automatically
0.30 to 0.69	Medium	Route to analyst review
0.70 and above	High	Block and investigate

Review record (blank template)

Field	Entry
Transaction ID	
Date and time reviewed	
Fraud probability (confidence score)	
Decision band	Low / Medium / High
Recommended action	Approve / Review / Block
Top contributing features	1. 2. 3.
Reason codes	
Analyst decision	Confirm fraud / Clear as legitimate / Escalate
Analyst notes	

Worked example (TXN_001)

Field	Entry
Transaction ID	TXN_001
Date and time reviewed	[example]

Field	Entry
Fraud probability (confidence score)	0.94
Decision band	High
Recommended action	Block and investigate
Top contributing features	1. 1-hour velocity = 7 transactions 2. IP risk score = 0.95 3. Account age = 5 days
Reason codes	VEL-1H, IP-RISK, ACCT-NEW
Analyst decision	Confirm fraud / Clear as legitimate / Escalate
Analyst notes	High confidence; multiple independent fraud signals present.

This template turns the model's SHAP output into a consistent review artifact. Each reviewed case records why the model flagged it, the features that drove the score, and the confidence behind it, which keeps human review fast, consistent, and auditable.